

Nitin Kumar Shukla



Professional Summary

AI Engineer (Integrated M.Tech, VIT 2027) specialising in LLM-powered applications, Retrieval-Augmented Generation (RAG), and scalable ML pipelines. Proficient in Python, LangChain, LangGraph, Semantic Kernel, and AWS. Experienced in automating enterprise workflows with NLP, OCR, and REST APIs. Proven track record of deploying production-grade AI solutions with measurable impact — including a 30% error reduction in a live medical-billing system.

Education

Vellore Institute of Technology (VIT)
Integrated M.Tech in Artificial Intelligence

2022 – 2027
CGPA: 8.14

Experience

AI Engineer Intern (Part-Time)
Factobi

May 2026 – Present
Remote

- Developing AI automation tools with .NET, C#, Semantic Kernel, and REST APIs deployed across enterprise workflows for multiple clients.
- Designing RAG pipelines using Pinecone/Chroma vector databases, cutting LLM hallucination rate by ~40% on domain-specific queries.
- Engineering multi-agent workflows with LangChain and LangGraph; reducing manual processing time by ~60% and improving task accuracy to 91%.

Data Analyst Intern

Oct 2024 – Dec 2024

South Eastern Coalfields Limited (SECL)

Bilaspur, India

- Automated reimbursement workflows for 500+ daily claims, reducing turnaround time by 45% via an end-to-end medical bill scrutiny system.
- Engineered OCR pipelines using Tesseract and OpenCV to extract structured data from medical records with 94% field-extraction accuracy; reduced processing errors by 30%.
- Designed Python anomaly detection modules to flag high-risk claims and integrated RESTful APIs for real-time data retrieval across the internal dashboard.

Projects

Deepfake Detection Platform | *TensorFlow, ResNet-50, OpenCV, Flask, MTCNN*

Sep 2023

- Architected a ResNet-50 CNN with custom attention layers to detect spatial inconsistencies and GAN-based artifacts via FFT preprocessing; integrated MTCNN for multi-face detection and alignment.
- Deployed a Flask REST API with OpenCV frame sampling; achieved 96.2% accuracy on FaceForensics++ and reduced inference latency by 38% via optimised batch prediction.

Stock Market Prediction System | *Python, XGBoost, Scikit-learn, Pandas, TA-Lib*

Jan 2024

- Designed an end-to-end ML pipeline on OHLCV data computing 20+ technical indicators (RSI, MACD, Bollinger Bands) with rolling-window lag features for temporal pattern extraction.
- Achieved F1-score of 0.83 on unseen market regimes via ensemble stacking (Random Forest + XGBoost), GridSearchCV tuning, and L1/L2 regularisation to reduce overfitting.

Technical Skills

Languages: Python (Advanced), Java, C++, JavaScript, SQL (PostgreSQL / MySQL)

AI / ML Frameworks: LangChain, LangGraph, Semantic Kernel, Hugging Face Transformers, TensorFlow, PyTorch, Scikit-learn, XGBoost, Pandas, TA-Lib, MLflow

AI / ML Concepts: Generative AI, Large Language Models (LLMs), RAG, Prompt Engineering, LLM Fine-Tuning, Natural Language Processing (NLP), Vector Databases, Multi-Agent Systems, Deep Learning

Cloud & DevOps: AWS (EC2, S3, Lambda), Docker, Kubernetes, CI/CD Pipelines, Git/GitHub, Linux Shell Scripting, Microservices

Core Concepts: Data Structures & Algorithms, System Design, REST APIs, FastAPI, SDLC, Agile/Scrum

Certifications

Kubernetes for Absolute Beginners | Packt | 2025

Applied Machine Learning in Python | University of Michigan (Coursera) | 2023